BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 7191 / 7198

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ-1. 1021 — (جماعت دہم) (Session 2017-19 to 2019-21) — Inter Part-II — Sargodha Board — (معروضی) ریاضی سائنس / Mathematics (Objective) — Group 1

Q. Time Allowed: 20 minutes — PAPER CODE 7191 — Maximum Marks: 15

Q1. The number of Terms in a standard quadratic equation $ax^2 + bx + c = 0$ is

(A) (B)

(C) (D)

Q2. If α, β are the roots of $3x^2 + 5x - 2 = 0$, then α + β is

(A) (B)

(C) (D)

Q3. The discrimination of $ax^2 + bx + c = 0$ is

(A) (B)

(C) (D)

Q4. If $y^2 \propto 1/x^3$, Then:

(A) (B)

(C) (D)

Q5. If a:b = x:y. Then alternando property is:

(A) (B)

(C) (D)

Q6. $(x^2+1)/((x-1)(x+2))$ is _____.

(A) (B)

(C) (D)

Q7. The different number of ways to describe a set are.

(A) (B)

(C) (D)

Q8. If A and B are disjoint sets, Then $A \cup B$ is equal to

(A) (B)

(C) (D)

Q9. In a cumulative frequency polygon, frequencies are plotted against.

(A) (B)

(C) (D)

Q10. The measure which determines the middle most observation in a data set is called

(A) (B)

(C) (D)

Q11. $\sec^2\theta$ = _____

(A) (B)

(C) (D)

Q12. Line segment joining any point of the circle to the centre is called:

(A) (B)

(C) (D)

Q13. Two Tangents drawn to a circle from a point outside it are of _____ in length.

(A) (B)

(C) (D)

Q14. A 4 cm long chord subtands a central angle of 60°. The radial segment of this circle is.

(A) (B)

(C) (D)

Q15. The measure of the external angle of a regular hexagon is

(A) (B)

(C) (D)

CAVEAT: the GI objective scan carries a heavy FGSTUDY.com watermark — option (A) of Q7 and Q14 is obscured by it and marked [illegible in source scan]; Q4 options (B) and (C)

and Q5 option (D) are too faint to read and marked the same. Board typos kept verbatim: 'The number of Terms', 'The discrimination of', 'Then:', 'alternando property is:', 'The different number of ways to describe a set are.', 'is called:', 'subtands', 'The measure of the external angle of a regular hexagon is'. The objective is printed in a bilingual table (Urdu + English); questions transcribed from the printed English column.

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QPart I. 1021 — (جماعت دہم) (Session 2017-19 to 2019-21) — Inter Part-II — Sargodha Board — (انشائیہ) ریاضی سائنس / Mathematics (Essay Type) — Group 1

Q. Time Allowed: 2:10 hours — Maximum Marks: 60

QPart II. Part II — حصہ دوم

Note: Attempt any Three Questions. Q.9 is compulsory. (8×3=24)

CAVEAT: the GI subjective scans are cut off at the left edge on Q3 parts (i), (ii), (v)–(ix) — the beginnings of those stems are physically missing from the source image and marked [illegible in source scan]. Board typos kept verbatim: 'Write the name of methods to solving a quadratic equation.', 'Find the sum and product of root.', 'How many minutes are in two right angles?', 'In ΔABC a = 17cm, b = 15 cm, c = 8 cm, m∠B = ?', 'For an arc draw two perpendicular bisectors of the chords', 'Solve the following equation using quadratic formula', "Solve by using synthetic division if '1' and '3' are the roots of the equation", 'Find the values of the letter involved in continued proportions', "then verify (A ∩ B)' = A' ∪ B'", 'a family bought 21.3 liters of petrol', 'Prove that: Any two angles in the same segment of a circle are equal.', 'Answer briefly any Six parts from the followings'.

GROUP – II

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ-1. 1021 — (جماعت دہم) (Session 2017-19 to 2019-21) — Inter Part-II — Sargodha Board — (معروضی) ریاضی سائنس / Mathematics (Objective) — Group 2

Q. Time Allowed: 20 minutes — PAPER CODE 7198 — Maximum Marks: 15

Q1. A circle has only one _____

(A) (B)

(C) (D)

Q2. The length of a chord and the radial segment of a circle are congruent. The central angle made by the chord will be

(A) (B)

(C) (D)

Q3. How many common tangents can be drawn for two touching circles.

(A) (B)

(C) (D)

Q4. Two linear factors of $x^2 - 15x + 56$ are

(A) (B)

(C) (D)

Q5. Roots of the equation $4x^2 - 5x + 2 = 0$ are

(A) (B)

(C) (D)

Q6. If $b^2 - 4ac > 0$, but not a perfect square then roots of $ax^2 + bx + c = 0$ are

(A) (B)

(C) (D)

Q7. In a ratio a:b, a is called

(A) (B)

(C) (D)

Q8. The fourth proportional w of x:y :: v:w is

(A) (B)

(C) (D)

Q9. A function of the form $f(x) = N(x)/D(x)$, with $D(x) \neq 0$, where N(x) and D(x) are polynomials in x is called.

(A) (B)

(C) (D)

Q10. Power set of an empty set is

(A) (B)

(C) (D)

Q11. If number of elements in set A is 3 and in set B is 2, then number of binary relations in $A \times B$ is

(A) (B)

(C) (D)

Q12. A data in the form of frequency distribution is called.

(A) (B)

(C) (D)

Q13. Mean is affected by change in

(A) (B)

(C) (D)

Q14. If tanθ = √3, then θ is equal to

(A) (B)

(C) (D)

Q15. A chord passing through the centre of a circle is called

(A) (B)

(C) (D)

CAVEAT: the GII objective scan is cut off at the left edge — options (D) of Q8, Q10, Q13, Q14, Q15 are physically missing from the source image and marked [illegible in source scan]. Board typos kept verbatim: 'Two linear factors of x^2 - 15x + 56 are', 'Roots of the equation $4x^2$ - 5x + 2 = 0 are', 'If b^2 - 4ac > 0, but not a perfect square then roots of ax^2 + bx + c = 0 are', 'In a ratio a:b, a is called', 'The fourth proportional w of x:y :: v:w is', 'A function of the form f(x) = N(x)/D(x), with D(x) ≠ 0, where N(x) and D(x) are polynomials in x is called.', 'Power set of an empty set is', 'If number of elements in set A is 3 and in set B is 2, then number of binary relations in A × B is', 'A data in the form of frequency distribution is called.', 'Mean is affected by change in', 'If tanθ = √3, then θ is equal to', 'A chord passing through the centre of a circle is called', 'How many common tangents can be drawn for two touching circles.' The objective is printed in a bilingual table (Urdu + English); questions transcribed from the printed English column.

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QPart I. 1021 — (جماعت دہم) (Session 2017-19 to 2019-21) — Inter Part-II — Sargodha Board — (انشائیہ) ریاضی سائنس / Mathematics (Essay Type) — Group 2

Q. Time Allowed: 2:10 hours — Maximum Marks: 60

QPart II. Part II — حصہ دوم

Note: Attempt any Three Questions. Q.9 is compulsory. (8×3=24)

CAVEAT: the GII subjective scans are cut off at the left edge on Q3 parts (i)–(vii), (ix) and Q9(a) — the beginnings of those stems are physically missing from the source image and marked [illegible in source scan]. Board typos kept verbatim: 'If the ratios 3x + 1 : 6 + 4x

and 2 : 5 are equal. Find value of x', 'Find the value of the letter involved in the continued proportion', 'If U = {1,2,3,4,...10}', "then verify (A - B)' = A' ∪ B", 'If tan θ = 4/3 and sin θ < 0, find the values of an other trigonometric functions at θ', 'Answer briefly any Six parts from the followings'.